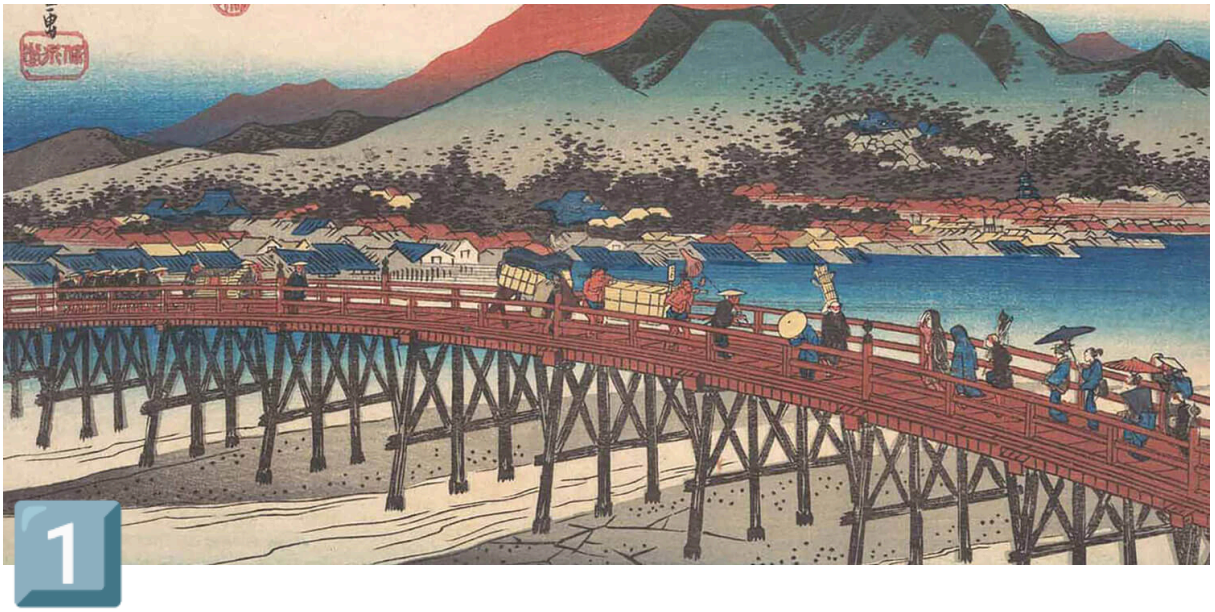
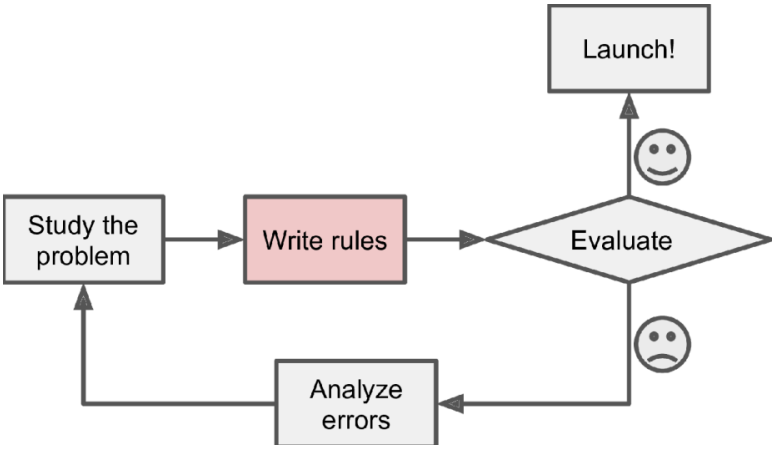


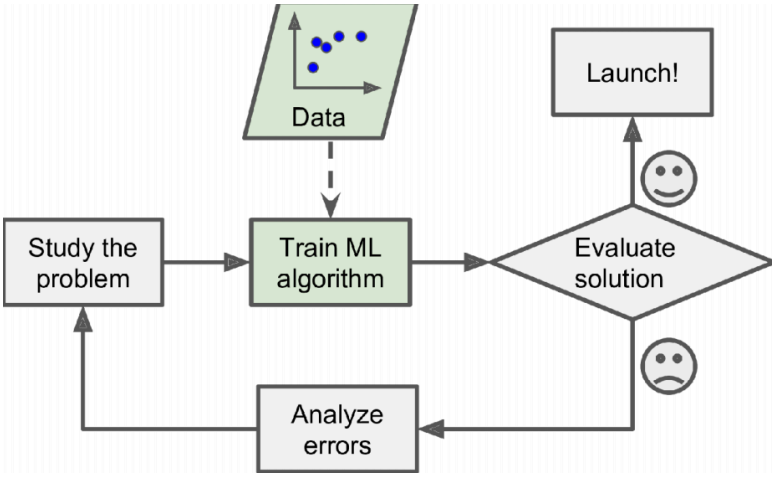
The Machine Learning Landscape

- Training Set: The Examples that the system use to learn.
- **Training instance:** Each training Example

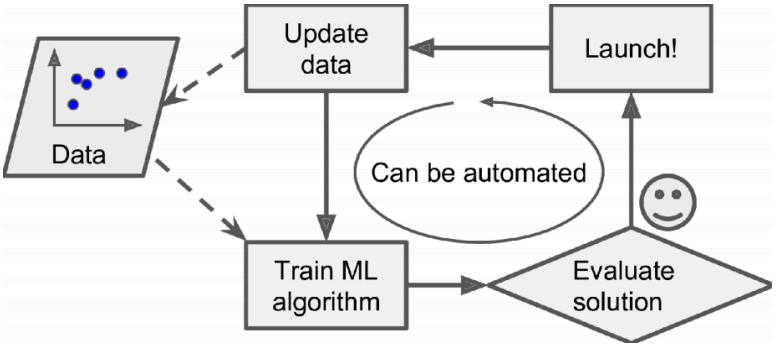
▼ 1. The traditional approach



▼ 2. Machine Learning Approach



▼ 3. Automatically adopting to change



🦋 Types of Machine Learning Systems:-

▼ 1. Supervised/Unsupervised Learning:-

- Classified into 4 types on the basis of amount and type of supervision they get during training.
- Dimensionality Reduction: Simplify a data without loosing too much inform.
- Feature Extraction: In case of interdependent features, we can merge them together as one.

▼ A. Supervised Learning:

- We feed a training set into an algorithm.
- Labels: The training set we feed into an algorithm.
- Classification: A typical Supervised learning
- Target: The value that is predicted

- Predicates: The set of features that is used to train a model.
- Some of the most important supervised learning algorithms are:
  - k-Nearest Neighbors
  - Linear Regression
  - Logistic Regression
  - Support Vector Machines (SVMs)
  - Decision Trees and Random Forest
  - Neural Networks

▼ **B. Unsupervised Learning:**

- There is no training data
- Some of the most important unsupervised learning algorithms are:
  - Clustering
    - K-Means
    - DBSCAN
    - Hierarchical Cluster Analysis (HCA)
  - Anomaly detection and novelty detection
    - One-class SVM
    - Isolation Forest
  - Visualization and dimensionality reduction
    - Principal Component Analysis (PCA)
    - Kernel PCA
    - Locally Linear Embedding (LLE)
    - t-Distributed Stochastic Neighbor Embedding (t-SNE)
  - Association rule learning
    - Apriori
    - Eclat

▼ **C. Semi-supervised Learning:**

- Mix of labeled and unlabeled instances

▼ **D. Reinforcement Learning:**

- The learning system, called an *agent* can observe the environment, select and perform actions, and get *rewards* in return (or *penalties* in the form of negative rewards.
- It must then learn by itself what is the best strategy, called a *policy*, to get the most reward over time.
- A policy defines what action the agent should choose when it is in a given situation.

▼ 2. Batch and Online Learning:

- Whether or not a system can learn incrementally from a stream of incoming data.
- 2 types are:

▼ **A. Batch Learning:**

- Incapable of learning incrementally.
- Trained using all available data.
- Offline Learning:- Once the model learns from a given set of data, it just predicts without learning anymore.
- If a Batch Learning System needs to learn from new data, entire dataset needs to be changed from scratch which is time consuming . 🥲

▼ **B. Online Learning:**

- Data instances are filled sequentially, either incrementally or in small group called *mini-batches.*
- Great for systems that receive data as a continuous flow.
- Updates its data, deletes old data if they remain unused.
- *Out-of-Core* learning: Used to train systems on huge datasets that cannot fit one machine's main memory.

- Learning Rate: How fast an online learning system adapt to changing data.
- **Cons or Hazards:**
  - If you have a higher learning rate, then system will rapidly adapt to new data but at the same time it forgets old data and vice versa (data is less sensitive to new data).
  - If bad data is fed to system, the system's performance will gradually decline.

▼ 3. Instance-based vs Model-based model:

- Depends on how a model predicts a data on unknown values.
- 2 types are:

  A. Instance-based Learning:

  - Filters data (ex.- spam) based on data fed into it and similar data.

  B. Model-based Leaning:

  - Build a model from train data and the model will make predictions.

🌿 Challenges in Machine Learning:-

▼ 1. Insufficient Quantity of Training Data

- Huge amount of data needed for machines to work properly.
- Complex problems (ex.- Image/Speech Recognition) needs millions of data.
-

▼ 2. Non-Representative Training Data

- Use a training set that is a representative of the model I want to generalize to.
- Sampling Noise: Happens when a sampling is too small, i.e., non-representative data as a result of chance.
- Sampling Bias: Happens when large samples can be non-representative as a result of chance.

▼ 3. Poor Quality Data

- If my training data is full of errors, outliers and noise, it will make it harder to system to detect the underlying patterns, so our system will less likely perform well.

▼ 4. Irrelevant Features

- System will be capable of learning from training data only if it contains only enough relevant data and not too many irrelevant data.
- Feature Engineering:
  - Feature Selection- Select the most useful features to train on among existing features.
  - Feature Extraction- Combining existing features to to produce a more useful one.
  - Create new features by gathering new data.

▼ 5. Overfitting the Training Data

- Overfitting- Model performs well on training data, but it doesn't generalize well.
- Do not over generalize the data.
- Regularization- Constructing a model to make it simpler and reduce the risk of overfitting.
- Hyper-parameter testing- Control of amount of regularization.

- Generalization Error- We train our model on train data and predict on test data. The error rate on new cases is Generalization Error.

▼ 🧮 Hyperparameter Tuning & Model Selection:-

- Overfitting- Happens when a model memorizes on training data, works well on it but does not learn the pattern. Hence the model becomes well trained on training data but poor on new data.
- Regularization- Used to reduce overfitting in ML. Instead of making the model too complicated, regularization penalizes large parameter values, forcing the model to stay simpler.

- Regularization Hyperparameter- Controls how strong the regularization parameter is.

> 💡 ***Regularization Cost Function:-***
> $J(\theta)$=Loss Function+λ·Regularization Term

- Holdout Validation- We hold a part of train dataset. Maybe we use 80% of the train data to train our model and use the rest 20% to test our data on and check its accuracy before predicting on test dataset.
- Validation Set- Portion of dataset that is used to tune our model and choose the best hyperparameters while training our data. It is separate from both test and train datasets.
    - Validation set is mainly used to choose best models, regularisation and other things before trying it out on real data. Its more like a mock test to practice from what we have learnt before giving a test.
- Cross Validation- We split a dataset multiple times into different combinations of train and validation set making our model more reliable.
- train-dev set- Detect if the problem with our model is due to is due to overfitting or due to data mismatch (test data differs from train data). If it performs well, the model is not overfitting the training set.